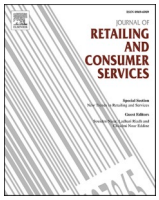
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Effects of human versus virtual human influencers on the appearance anxiety of social media users

Fengyi Deng *, Xia Jiang

Chongqing Technology and Business University, Chongqing, China

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ABSTRACT

The association between social media use and an individual's appearance dissatisfaction has generated broad scholarly interests. This study explored the specific appearance dissatisfaction resulting from the effects of social media influencer images on the appearance anxiety of users, in particular the differences between human influencers (HIs) and emerging virtual influencers (VIs). Women (n = 178) aged 18–35 were randomly assigned to HI images, VI images, and scenic images (control group) and were asked to answer related questionnaires. The results showed that images of both HI and VI induced significantly higher appearance anxiety than in the control group. The participants who were exposed to VI images reported significantly lower appearance anxiety than those who were exposed to HI images. Participants' state appearance comparison played a mediating role between the images of social media influencers and appearance anxiety. In addition, participants who viewed VI images reported significantly lower state appearance comparison than those who viewed HI images. Insights for influencer marketing practice were further discussed based on the results.

1. Introduction

With the rapid development of social media, the dark side of influencer marketing has recently come under the academic spotlight (e.g. Seek and Barker, 2022; Naeem, 2021). Evidence from a meta-analysis of 24 experimental studies revealed that images of the ideal appearance depicted on social media have a moderate negative effect on body image (Hedges' $g = -0.61$, $p < 0.01$; Valle et al., 2021). Appearance anxiety, which refers to a function of responses (e,g, anxiety, tension, or nervousness) to body image, and shame, which includes various cognitive processes, are among the important indicators of body images which have gained interest among many scholars (Cihan and Cihan, 2019; Velkoff and Smith, 2020). Women who reported appearance anxiety are more likely to see themselves as objects (state self-objectification) to be viewed and evaluated based on their appearance, causing a series of mental health risks (e.g., sensitive feelings, negative mood, depression) (Dimas et al., 2021; Velkoff and Smith, 2020). Many scholars have explore the role of social media in eliciting people's appearance anxiety and have argued that social media may have a significantly greater impact on appearance-related concerns than traditional media (Hawes et al., 2020). The user-generated content (UGC) and real-time feedback features on social media form a complex system which not only enables individuals to judge others' and their own appearance, but also serves to reinforce appearance ideals where they indicate social approval of those ideals. (Valle et al., 2021; Hawes et al., 2020).

Previous research attention has shifted from discussing the effects of social media use (e.g., Fardouly and Vartanian, 2016) to investigating those of social media content (e.g., Slater et al., 2019; Rodgers et al., 2021). However, the role of societal beauty standards has been overlooked. The visual presentations of influencers, friends, and strangers, as well as the comments they receive, cultivate social stereotypes and societal beauty standards (Saunders and Eaton, 2018; Valle et al., 2021). The societal beauty standards conveyed through social media are more influential than those delivered through traditional media, as people tend to believe that social media reflect the reality surrounding them (Fardouly and Vartanian, 2016).

In the above mechanisms, social media influencers (SMIs) play an important role as key opinion leaders regarding societal standards (Ge and Gretzel, 2018). The term SMIs refers to people who have "gained popularity due to their social media presence and content" (Wahab et al., 2022; Aw and Chuah, 2021). Although many studies have identified the association between social media and appearance anxiety, none has highlighted the role of SMIs directly. The typical experimental

* Corresponding author.
E-mail addresses: secilia.deng@ctbu.edu.cn (F. Deng), jiangxiaecho@ctbu.edu.cn (X. Jiang).

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operation in existing studies involves asking participants to view ideal and non-ideal appearance images from social media (e.g., Carter and Vartanian, 2022). Such studies do not specifically state whether the images used are of SMIs. Therefore, given this research gap, the question of whether SMI images of elicit the appearance anxiety among social media users needs further discussion.

Moreover, with continuous development in science and technology, use of new SMI forms has been highlighted as a valuable emerging marketing strategy (Farivar and Wang, 2022; Bu et al., 2022; Cheung et al., 2022; Appel et al., 2020). Virtual influencers (VIs), who are designed to avoid the limitations of real humans, are believed to have a more prominent presence than human influencers on social media, along with strong computing power and artificial intelligence (AI) algorithms (Appel et al., 2020). Defined as "embodied virtual agents" (Tan et al., 2020), several VIs have an anthropomorphized appearance and are able to share attractive content, especially visual content, on social media (Arsenyan and Mirowska, 2021). Most VIs are built on their own story lines and have distinct personalities. The interest generated by these VIs among social media users is an aspect that needs further research. For example, millions of followers pay attention to the daily fashion choices and routines of Lil Miquela, the first computer-generated imagery (CGI) in the world. As VIs gain increasingly prominent social and marketing value, the influential mechanism of VIs has generated interest among many scholars (Arsenyan and Mirowska, 2021; Faddoul and Chatterjee, 2020). Past studies have focused on people's reactions to VIs and human influencers (HIs) on social media and investiagted which VIs' characteristics that may impact people's perceptions (e.g., Chiou et al., 2019; Schroeder et al., 2020). Based on the above discussion, the present study aims to examine whether images of HIs and VIs induce social media users' appearance anxiety and whether there is a significant difference between the effects of HIs and VIs.

Considering the growing number of studies of gender differences in the relationships among social media and appearance-related outcomes (e.g., Rogers and Rosseau, 2022), the present study focuses on the impact of SMI images on young females as this group has been found to a higher level of self-objectication, appearance-related motivations, and social media appearance preoccupation (e.g., Zimmer-Gembeck et al., 2021; Rodgers et al., 2020; Choukas-Bradley et al., 2022).

1.1. The effect of SMI images on social media users' appearance anxiety

Extensive experiments and surveys have been conducted to confirm the relationship between exposure to ideal imagery and appearance anxiety (e.g., Slater et al., 2019). In this mechanism, an appearance-related photo is associated with appearance comparison, which, in turn, elicits body dissatisfaction (Lee and Lee, 2021). It has been claimed that celebrities are particularly harmful targets of appearance comparison due to their attractive appearance (Fardouly et al., 2021). On social media, most HIs meet the general social appearance ideals (i.e., thin, fair-complexioned, and young; Valle et al., 2021). Their attractiveness and appearance characteristics have been demonstrated to be the factors that influence social media users the most (Fernandes et al., 2022; Sokolova and Kefi, 2019). Therefore, the first hypothesis of this study is:

Hypothesis 1. HI images have a positive impact on an individual's appearance anxiety.

However, no research has identified the direct association between VIs' appearance and social media users' appearance anxiety. Researchers have demostrated the impact of virtual people's appearance on individuals' attitudes and behaviors (e.g., Hanus and Fox, 2015). Visually attractive VIs are more likely to elicit social responses and behavioral change in humans (Arsenyan and Mirowska, 2021). As it is their conformity with attractive stereotypes which gives VIs their persuasive power and effectiveness (Faddoul and Chatterjee, 2020), many VIs are designed to present an idealized appearance. Based on the research results that ideal body pictures can elicit appearance anxiety (Slater et al., 2019), Hypothesis 2 is formulated:

Hypothesis 2. Images of VIs have a positive impact on individuals' appearance anxiety.

In the discussion of whether people react differently according to an influencer's nature (human vs. virtual), several researchers have claimed that individuals' cognition, emotion, behavior, and even related brain activation regions present the same paradigm when encountering both types (i.e., VIs and HIs) (e.g., Stein et al., 2022). Social presence plays a significant role in the anthropomorphic interactions investigated in such studies (Potdevin et al., 2020). However, evidence from other studies has identified the differences between the influence of computers and humans. People tend to tolerate emotional quirks when they face a computer-generated partner and have greater involvement in conversations with human partners (Arsenyan and Mirowska, 2021). Furthermore, people also had negative perceptions of VIs' human-resembling abilities (MacDorman, 2019; Stein et al., 2020). Most of these studies were dependent on the uncanny valley (UV) theory (Mousas et al., 2018), which describes the decreased affinity toward a virtual person who increasingly resembles a human. Considering such contradictory conclusions, the following research question (RQ) was formulated:

RQ1. Is the effect of VI images different from that of HI images on individuals' appearance anxiety?

1.2. The mediating role of state appearance comparison

Comparison is significant in the mechanisms underlying the effects of social media on individuals (Valle et al., 2021; Fardouly et al., 2017; Hawes et al., 2020). Decades ago, researchers noticed that the sharing of an ideal life and a positive self-image on social media may induce negative feelings (such as jealousy) among others (Tandoc et al., 2015). Further research demonstrated that social comparison can help in understanding this phenomenon (Saunders and Eaton, 2018), in particular upward comparisons, which are characterized by comparing oneself to someone deemed as more attractive or superior in some way (McComb and Mills, 2022). According to the social comparison theory, individuals evaluate their social value by comparing their performances and characteristics in various aspects (such as finance, appearance, and intelligence) with those of others. Additionally, individuals are more likely to perceive those who are similar to themselves as social comparison targets. (Fardouly et al., 2017). As most of the content is generated by friends and peers, social media produces a higher level of perceived proximity than traditional media (Rodgers et al., 2021). The platforms provided by social media are also highly accessible and visual, facilitating in social comparison in a relatively easy manner (Fardouly et al., 2017). Consequently, people may tend to compare themselves with figures on social media.

Moreover, most images on social media are edited by modern media technology (Cohen et al., 2018). User-friendly photo-editing tools meet a user's desire to perfect their preferred self-image. Thus, exposure to these curated images on social media, which are closer to the idealized societal appearance, become upward comparison targets for social media users (Anixiadis et al., 2019; Fardouly and Rapee, 2019). Therefore, stress, depression, and anxiety are triggered and intensified, when users realize that their appearance cannot meet the societal beauty standards (Saunders and Eaton, 2018; Tiggemann and Zinoviev, 2019). Studies have been conducted to identify the above-mentioned mechanism. Researchers have found that comparison plays an important mediating role in social media's effect on individuals' appearance-related concerns (e.g., Carter and Vartanian, 2022; Boursier et al., 2020; Slater et al., 2019). The academic concept, state appearance comparison, has been used in these studies to explain the extent to which the participants engaged in appearance comparison processing while viewing the images (e.g., McComb and Mills, 2022). State measure has

been identified as a useful tool to validate the power of experimental manipulations when specific image disturbances are used (Reed et al., 1991). Therefore, we speculate that individuals' state appearance comparison may function as a mediator between SMI images and appearance anxiety. Thus, the following hypothesis has been formulated:

Hypothesis 3. Appearance comparison mediates the effect between the images of influencers (both HI and VI) and individuals' appearance anxiety.

According to the above discussion of whether people react differently depending on an influencer's nature (human vs. virtual), we formulate RQ2 as follows:

RQ2. Do images of VIs and HIs affect individuals' state appearance comparison differently?

Fig. 1 shows the research model of this study.

2. Method

2.1. Experiment design

A single-factor inter-group experiment was conducted in this study. Three (HI group vs. VI group vs. control group) levels were created. The participants were randomly assigned to the above groups evenly, and they reviewed the corresponding images.

2.2. Participants

The participants (n = 202) were recruited through Credamo, an Internet questionnaire platform in China, and were paid 5 RMB each for participation. Credamo contains a database of 2.8 million national samples. Referring to the operation from Mulgrew and his colleagues' study (2017), only 18–35-year-old (the age group most concerned about their appearance) females were eligible to participate in this experiment. The responses were excluded if the participants failed to answer two attention check questions (n = 5), spent too much or little time on the experiment (n = 14), or chose the same answer for all the measures (n = 5). Finally, 178 respondents were remained (60 for the VI group, 58 for the HI group, and 60 for the control group), with a mean age of 26.24 (SD = 4.82) and a mean body mass index (BMI) of 19.71 (SD = 3.18).

2.3. Materials

2.3.1. Stimuli materials

Three sets of 10 images were used in the experiment. The first set comprised images of Ayayi, the three-dimensional (3D) realistic virtual character created in 2021 by a Chinese Internet company. Ayayi shares her wardrobe and makeup style realistically and energetically through social media. At present, Ayayi is at the initial stage of becoming a virtual fashion influencer, and she was selected for this study for two principal reasons.First, as a virtual fashion blogger Ayayi is designed to have a young, slim, toned body, and an exquisite appearance. According to previous studies, these images have the highest impact in terms of inducing individuals' body dissatisfaction (Valle et al., 2021; Cohen et al., 2019). Second, Farivar and his colleagues (2021) claimed that body satisfaction is greater when women are exposed to a favorite celebrity with whom they have a para-social relationship, and Brown and Tiggemann (2016) believed that high celebrity worship worsens body dissatisfaction after exposure to thin ideal celebrity images. As a newly-born VI, Ayayi has fewer than 60,000 followers (compared to the 3 million following Miquela, another VI). Referring to McComb and Mills' study (2022), Ayayi was selected to minimize the impact of celebrity worship and para-social relationships on appearance anxiety. The second set of images was selected from an HI with almost the same haircut, body shape, and number of followers as Ayayi (see Supplementary Materials for sample images). Ten images each were selected from both Ayayi's and the HI's websites, including full body shots and close-ups of their faces. No other people were shown in those images. The third set of images comprised scenic pictures including mountains, the ocean, and gardens.

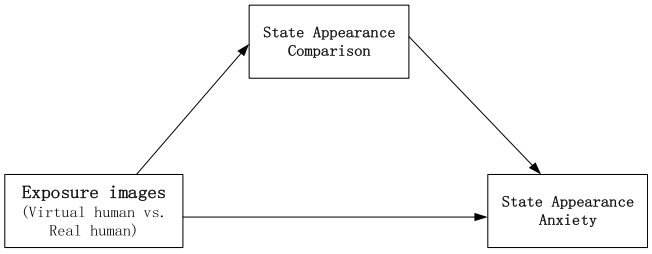


Fig. 1. Research model.

Physical attractiveness was examined on a seven-point Likert scale, as in a previous study (Li and Peng, 2021). The scale consists of four items and asks participants whether they feel that the woman in the picture is classy, beautiful, elegant, and sexy. As a pretest, 60 female participants aged 18–35 were recruited and divided evenly into two groups. Each group was shown one set of images. An independent samples *t*-test showed no significant difference ($M_{HI} = 5.70$, SD = 0.82; $M_{VI} = 5.50$, SD = 0.99; t = 0.858, $p > 0.05$, df = 58) in attractiveness between the two sets of images.

2.3.2. Measures

2.3.2.1. State appearance anxiety. The measurement of a participant's appearance anxiety was based on the state subscale of the physical appearance state-trait anxiety scale (PASTA-S) developed by Reed et al. (1991). The PASTA-S measures a person's levels of anxiety, tension, or nervousness regarding their own appearance and specific body parts "right now" for 16 body parts (hips, tights, stomach, arms, waist, neck, chin, feet, legs, forehead, skin, curve, weight, height, wrists, and buttocks) on a five-point Likert scale (1 = not at all, 2 = slightly, 3 = moderately, 4 = very much so, and 5 = exceptionally so). The PASTA has been widely used in research on body image (e.g., Bue, 2020; Porras--Garcia et al., 2020). The reliability of the scale was good (α = 0.909), and the total score of the 16 items represented the extent of each participant's appearance anxiety, with a higher score indicating a higher level of state appearance anxiety.

2.3.2.2. State appearance comparison. The state appearance comparison scale developed by Tiggemann and McGill (2004) was adapted to measure the level at which the participants compared their appearance with that of the individuals in the images. A four-item, seven-point Likert scale (from 1 = not at all to 7 = very much) was used. The first two items asked the participants to evaluate the level at which they thought about (1) their whole appearance and (2) specific parts of their bodies while observing the person in the images. The last two items asked the participants to evaluate the level at which they compared (3) their whole appearance and (4) specific body parts with those of the person in the images. The total score of the four items represents the extent of each participant's appearance comparison, with a higher score indicating a higher level of comparison with the person in the images (α = 0.925).

2.3.2.3. Covariates. Social media use (e.g., time and frequency) has proven to have significant effects on individuals' appearance anxiety (Hawes et al., 2020; Phillips and Wisniewski, 2021). Moreover, BMI (Carter and Vartanian, 2022) and para-social relationships (Farivar et al., 2021) have been correlated with state body dissatisfaction. Thus, these variables were measured as covariates in the following analysis process. As regards, the social media use, the participants were asked the

average time they spent on social media every day (<1 h, 1–3 h, 3–5 h, >5 h), and the frequency of their social media use per day (<10 times, 10–20 times, 20–30 times, >30 times). The value of participants' BMI was calculated by dividing their weight (kg) by the square of their height (m). In addition, the participants were asked whether they were already aware of the influencer,

2.4. Procedure

To not disclose the real purpose of this experiment, the researchers asked the participants to judge whether the selected influencer was suitable as a social media spokesperson for a makeup brand for a specific consumer group. First, the participants were asked to provide their personal information and social media usage situation. After being given a brief written introduction about the definitions of a VI (VI group) and HI (HI group), the participants were asked to view each image for at least 10 s and answer whether the person in them was suitable to endorse a makeup brand. Next, the participants had to finish the state appearance anxiety scale, the state appearance comparison scale, and the attractiveness scale. Participants in the control group were asked to view 10 landscape images (for at least 10 s each) and then finish the state appearance anxiety scale and social media use situation items. The participants were rewarded 5 RMB after the researchers checked their responses.

2.5. Data analyses

Several checks were conducted prior to the main analysis process. First, an independent samples *t*-test showed that there is no significant difference in attractiveness ($M_{HI} = 6.07$, SD = 0.61; $M_{VI} = 5.77$, SD = 1.10; $t = -1.843$, $p > 0.05$, $df = 116$) between the HI and the VI group. Second, a one-way ANOVA analysis showed that there is no difference among the three exposure groups in terms of age [$F(2,175) = 1.568$, $p > 0.05$], average time spent on social media every day [$F(2,175) = 1.752$, $p > 0.05$], frequency of social media use per day [$F(2,175) = 1.234$, $p > 0.05$], or BMI [$F(2,175) = 0.532$, $p > 0.05$]. For the following analysis, the control condition was coded as 0, VI condition was coded as 1, and HI condition was coded as 2. Independent samples *t*-test analyses were conducted to test Hypotheses 1 and 2, and to answer RQ1 and 2. In addition, the SPSS PROCESS was conducted to test Hypothesis 3. Bootstrapping analysis with 5000 samples (Model 4; Hayes, 2017) was used to examine the mediation effect of the state appearance comparison.

3. Results

A one-way ANOVA was conducted to test the significance of main effect. State appearance anxiety [$F(2,175) = 9.289$, $p < 0.001$] and state appearance comparison [$F(2,175) = 5.078$, $p < 0.05$] were significantly different among exposure groups. According to the results of the independent samples *t*-test shown in Table 1, participants exposed to HI images reported a higher level of state appearance anxiety than those exposed to scenic images ($M_{HI} = 39.76$, SD = 11.18; $M_{Control} = 31.20$, SD = 9.85; $t = -4.408$, $p < 0.05$, $df = 116$). Thus, Hypothesis 1 was supported. Participants who viewed VI images also reported a higher level of state appearance anxiety than those who viewed control images ($M_{VI} = 35.15$, SD = 11.31; $M_{Control} = 31.20$, SD = 9.85; $t = -2.040$, $p < 0.05$, $df = 118$). Therefore, Hypothesis 2 was supported. In addition, the results indicate that participants in the HI condition had higher appearance anxiety, on average, than those in the VI condition ($M_{HI} = 39.76$, SD = 11.18; $M_{VI} = 35.15$, SD = 11.31; $t = 2.226$, $p < 0.05$, $df = 118$). Thus, RQ1 was answered.

Table 1
The Results of Independent Samples *t*-test.

Dependent	Images	M	SD	t	p	df
State Appearance Anxiety	HI	39.76	11.18			
	Control	31.20	9.85	−4.408	0.000	116
	VI	35.15	11.31			
	Control	31.20	9.85	−2.040	0.044	118
	HI	39.76	11.18			
	VI	35.15	11.31	2.226	0.028	116
State Appearance Comparison	HI	20.24	5.85			
	VI	17.67	6.53	2.258	0.026	116

Notes: HI means human influencer, VI means virtual influencer.

The mediating effect of the state appearance comparison was examined through bootstrapping analysis from the SPSS PROCESS. The data from the HI and VI group were analyzed with 5000 samples (Model 4; Hayes, 2017). BMI, frequency of social media use per day, and average time spent on social media every day were also included in this model as covariates, along with whether the participants were previously aware of the influencer. As shown in Table 2, the total effect of influencers' images on participants' appearance anxiety was significant (b = 4.532, SE = 2.116, 95% CI [0.340, 8.724]). In addition, the indirect effect of image exposure on the participants' appearance anxiety through the state appearance comparison was significant (b = 2.787, SE = 1.119, 95% CI [0.837, 5.250]). However, the direct effect of influencers' images on participants' appearance anxiety was not significant (b = 1.744, SE = 1.940, 95% CI [−2.101, 5.589]). Thus, Hypothesis 3 was supported. An individual's state comparison played a total mediating role between exposure to the SMI images and their state appearance anxiety. The positive coefficients indicated that individuals who engaged in greater appearance comparison experienced greater appearance anxiety levels.

Based on the independent samples *t*-test, participants who viewed HI images engaged in a higher level of state appearance comparison than those who viewed VI images ($M_{HI} = 20.24$, SD = 5.85; $M_{VI} = 17.67$, SD = 6.53; $t = 2.258$, $p < 0.05$, $df = 116$). Therefore, RQ2 was answered. HIs elicited a significantly higher state appearance comparison than VIs.

4. Discussion

4.1. Conclusions and implications

Based on the above analysis, we concluded from the present study that 1) the randomization of the experiment procedure resulted in no differences among HI, VI, and scenic/control groups on the covariates; 2) the groups of participants who viewed the images of VIs and HIs both reported significantly higher scores in appearance anxiety than the group which viewed scenic images; 3) participants who viewed VI's images reported significantly lower scores in state appearance anxiety than those who viewed HI images; 4) the level of state appearance comparison elicited by VIs was significantly lower than that elicited by HIs; and 5) individuals' state appearance comparison plays a mediating role in the association between an influencer's images and the individual's appearance anxiety. These findings provide evidence for the correlation between influencers' images and social media users' appearance anxiety. Contrary to previous works claiming that individuals' emotional reaction when encountering VIs present the same paradigm as that when encountering real HIs (e.g., Kramer et al., 2015), this study argued that the level of social media users' appearance anxiety varies with the nature of the influencers (real human vs. virtual human).

Table 2
Total, direct, and indirect effects of images on appearance anxiety.

Effect	Coefficient	SE	95% Bootstrap CI	
			Lower	Upper
Total	4.532	2.116	0.340	8.724
Direct	1.744	1.940	−2.101	5.589
Indirect	2.787	1.119	0.837	5.250

The findings of this study have several implications for influencer endorsement marketing practice. First, with the rapid development of SMI endorsement marketing, our findings have implications for practitioners using SMIs for marketing. The effect of human influencers on consumers' appearance anxiety should be considered when marketers specify strategies to stimulate consumers' purchase intention online. For instance, the effect of an HI's endorsement of cosmetic products may be better than that of a VI's endorsement, because the appearance anxiety aroused by SMIs may cause a stronger impulse among consumers to buy compensation products.

Second, the application of VI might be a useful strategy to protect individuals from developing negative appearance concerns due to social media use (e.g., Song et al., 2014; Giorgianni et al., 2020). Even though the distance between virtual people and real humans is increasingly reducing due to the rapid development of AI technology, it may be advisable for practitioners to retain certain elements of "virtuality" while designing a virtual person. Researchers have identified that people are more likely to compare themselves with individuals who are similar to them (Trekels et al., 2018). Maintaining the difference between VIs and real humans can reduce the appearance comparison between social media users and VIs, thus control ling the appearance anxiety level of users.

However, when a VI is used as the spokesperson of brands and products, we must also take into account that they are not fully identifiable as real people. The product category of VI endorsement needs careful consideration, because a virtual human is not as effective as a real human in specific ways. For instance, we believe that virtual people can "see" the world like real people but cannot "taste" food like real people.

4.2. Limitations

This study inevitably has some limitations. Most previous studies used images of several different people from social media (Valle et al., 2021). However, by limiting this study to the number of followers, only one influencer was selected for each group. This may have affected the validity of the experiment, or may have increased the impact of the influencer herself. Second, the effects of variables such as a para-social relationships and celebrity worship were not perfectly avoided by controlling the number of followers. Some participants were aware of the influencers selected in this study. Therefore, if sufficient research funds are available, the ideal research design would be to invite the technology company involved to create a brand-new VI. Third, it is difficult when using an online research platform such as Credamo to contact participants for further interview. Future research may consider the combination of online and offline experiments.

4.3. Future research directions

Considering the increasing number of studies on VIs' impact on social media users, an increasing amount of research should be conducted to gain further understanding of this mechanism. One direction includes the characteristics of VIs, for instance, whether people's perception of the human resemblance of VIs has an impact on the latter's influence (Pelau et al., 2021). Human resemblance was earlier defined as an individual's perception of humanistic features in objects (Pelau et al., 2021), and it is believed to be a significant variable that can affect individuals' reaction to AI-associated technologies, including VIs (Arsenyan and Mirowska, 2021; Kim et al., 2022). The boundary between HIs and VIs is becoming increasingly blurred. As VIs increasingly resemble real humans, further research is required to determine whether they will induce the same appearance anxiety as HIs do. After all, according to the UV hypothesis (Arsenyan and Mirowska, 2021), a VI being particularly similar to a real person will lead to an extreme emotional aversion among humans.

The second direction involves the discussion of the effects of users' characteristics. To date, research has shown that the appearance comparison tendency (Politte-Corn and Fardouly, 2020) and self-concept clarity (Carter and Vartanian, 2022), of individuals play a moderating role in the relation between social media exposure and the state appearance comparison. However, the moderating effect of these variables were not significant in this study's pretest experiments (not reported in the article due to their non-significant effect). Thus, research on more possible moderating variables is needed. For example, it has been proposed that media literacy skills have a protective effect by minimizing social media's negative impact on individuals' appearance anxiety (Rodgers et al., 2021; Tamplin et al., 2018). Whether media literacy skills play a moderating role in the context of new media technology, such as virtual figures based on AI, needs further research.

Third, the emotional relationship between consumers and VIs, such as worship and para-social relationships, deserve further discussion (Farivar et al., 2021). Based on the results of our experiment, the participants' previous awareness of this influencer ($b = 3.532$, $SE = 1.147$, 95% CI [1.258, 5.805]) was also identified to have a positive effect on their state appearance comparison. Future research can further explore how these emotional relationships affect consumers' psychological variables and purchase intentions.

Public comments on social media also play an important role in marketing and social media's effect on individuals' body images (Arsenyan and Mirowska, 2021; Hawes et al., 2020; Valle et al., 2021). Arsenyan and Mirowska (2021) predicted that the ideal images of SMIs might be more damaging to individuals' body images in a higher-risk one (comments praising the person's appearance) than a lower-risk context (comments highlighting the unrealistic nature of the images). Future studies could examine the role of other users' comments in the image-anxiety mechanism. Further, different media forms (words, pictures, and videos) may have different effects on individuals' appearance anxiety. Research involving in-depth discussion of the differences among text-based media (e.g., Twitter), picture-based media (e.g., Instagram), and short video-based media (e.g., TikTok) is needed in the future.

Data availability

Data will be made available on request.

Appendix B. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jretconser.2022.103233.

Appendix A. Part of the Experiment Stimuli

Part of VI images:



Part of HI images:



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